Question 8: Planning and Roadmap: Reading Validation Component - Q2 2026 Implementation Plan

Component: M2: Validate Reading Use Case

Timeline: April 1 - June 30, 2026 (12 weeks)

Go-Live: July 1, 2026

Owner: Lead Systems Analyst

Objective (Quarter Goal)

Deliver a production-ready reading validation component that processes 48M readings per day with 99.9% accuracy, enabling automated validation for 95% of volume while providing manual workflows for exceptions.

Key Themes

1. Revenue Protection

- Zero data loss at 48M readings/day
- Accurate billing calculations (temporal pricing)
- Complete audit trail (7-year retention)

2. Operational Efficiency

- Automated validation for 95% of readings
- Manual approval workflow for edge cases (5%)
- Field technician correction flow (0.1%)

3. Production Scalability

- Sustain 1,500 readings/sec (3× safety margin)
- Database partitioning from day one
- Graceful degradation under load

Quarter Timeline

Month 1: Planning & Design (Weeks 1-4)

Weeks 1-2: Foundation

- Database schema design (5 core tables)
- API specification and contracts
- Infrastructure provisioning (PostgreSQL + TimescaleDB)
- CI/CD pipeline setup

Weeks 3-4: Core Validation

- Reading ingestion API (1,000 readings/sec target)
- Automated validation engine (duplicate detection, business rules)
- Age-based routing (0-30 auto, 31-90 manual, 90+ reject)

Key Deliverables:

- ✅ Schema peer-reviewed and approved
- ✅ 1,000 readings/sec sustained with p95 <200ms
- ✅ Monitoring dashboards operational

Month 2: Build & Integration (Weeks 5-8)

Weeks 5-6: Manual Approval Workflow

- Supervisor work queue UI
- Approve/reject actions with reason capture
- SLA tracking (3-day approval window)
- Email notifications for escalations

Weeks 7-8: Field Tech Correction Flow

- Error investigation dashboard
- Correction submission form
- Version chaining (ERROR → corrected reading)
- Immutability enforcement (original ERROR never changes)

Key Deliverables:

- ✅ Supervisor UAT sign-off (8/10 satisfaction)
- ✅ Field Tech UAT sign-off (8/10 satisfaction)
- ✅ 100% audit trail on all corrections

Month 3: Testing & Release (Weeks 9-12)

Weeks 9-10: Performance & Scale Testing

- Load testing (1,500 readings/sec for 1 hour)
- Chaos testing (pod failures, network partitions)
- Database performance tuning
- Cost optimization (infrastructure right-sizing)

Week 11: Integration & UAT

- Smart meter integration (100 real meters)
- Billing system integration testing
- End-to-end UAT with all stakeholders
- Production deployment rehearsal

Week 12: Production Deployment

- July 1 go-live (zero downtime deployment)
- Smoke tests (first 1M readings)
- 48-hour hypercare (war room support)
- Handover to operations team

Key Deliverables:

- ✅ 1,500 readings/sec sustained (p95 <200ms, p99 <500ms)
- ✅ Zero P1/P2 bugs in production
- ✅ Operations confident to run independently

Success Metrics

Metric	Target	Measurement
Transaction success rate	≥99.9%	Readings successfully validated / total received
Throughput	1,500 readings/sec	p95 latency <200ms during peak load
Manual approval SLA	<3 days average	Time from PENDING_APPROVAL to decision
Data loss	0%	No readings lost during processing
Operations adoption	90%	Supervisors/Techs trained and using system

Prioritization & Reasoning

1. Foundation First (Weeks 1-2) - Why?

Database schema must be correct before building on it. Fixing schema with billions of rows in production is expensive and risky. **2 weeks upfront saves months of rework.**

2. Automated Path Before Manual (Weeks 3-4) - Why?

95% of volume (45.6M readings/day) flows through automated validation. This is the **revenue-critical path**. Manual workflows are lower volume and can tolerate early iteration.

3. Manual Workflows Second (Weeks 5-8) - Why?

Depends on core validation working. Lower volume means we can **refine based on user feedback**. Supervisor and Field Tech UIs benefit from real-world testing.

4. Performance Testing Before Go-Live (Weeks 9-10) - Why?

Cannot risk discovering bottlenecks in production. Testing at **3× target capacity** ensures safety margin. Finding issues in Week 10 vs Week 13 is the difference between fixing vs firefighting.

Risks & Mitigation

Risk	Probability	Impact	Mitigation
Database performance degrades at scale	Medium	High	Week 9-10 load testing with 3× volume; TimescaleDB partitioning from day 1
Supervisor UI too complex	Medium	Medium	Weekly demos to supervisors (Weeks 5-7); iterative feedback
Billing integration delayed	Low	High	Mock API for parallel development; dedicated

			integration track Week 8-11
Scope creep (new features mid-sprint)	High	Medium	Strict MVP definition; change control board (Lead SA + PO approval required)
Load shedding disrupts go-live	Medium	High	Deploy outside load shedding windows; 48-hour generator capacity at DC

Team Structure

Role	FTE	Responsibility
Lead Systems Analyst	1.0	Design, requirements, stakeholder alignment
Product Owner	0.5	Backlog, UAT sign-off
Tech Lead	1.0	Architecture, code review
Backend Engineers	2.0	Validation service, APIs
Frontend Engineer	1.0	Supervisor/Tech UIs
DBA	1.0	Schema, performance tuning
QA Engineer	1.0	Test automation, UAT
DevOps Engineer	1.0	Infrastructure, CI/CD

Total: 7.5 FTE × 3 months = R 2.73M budget

Dependencies

Dependency	Required By	Owner	Status
Infrastructure (AWS, databases)	Week 1	DevOps	Pre-provisioned in Week 0
SSO/Authentication	Week 2	Identity Team	Mock auth Weeks 2-4, real SSO Week 5
Test meters (100 real units)	Week 11	Operations	Simulated data until Week 11
Billing system API spec	Week 10	Billing Team	Spec agreed by Week 8

What's NOT in MVP (Post-Q2)

Deferred to Q3 2026:

- Machine learning anomaly detection (needs 3-6 months baseline data)
- Mobile app for field techs (convenience, not critical)
- Bulk approval operations (optimize after observing usage patterns)
- Advanced analytics dashboards (build on real production data)

Deferred to Q4 2026:

- Retroactive recalculation (P3 use case - separate project)
- Payment gateway integration (separate domain)

Key Milestones

Week	Milestone	Exit Criteria
2	Schema Complete	Schema peer-reviewed, indexes optimized, partitioning implemented
4	Auto Validation Works	1,000 readings/sec, p95 <200ms, 99.9% accuracy

6	Manual Approval Live	Supervisor UAT sign-off, SLA tracking operational
8	Correction Flow Complete	Field Tech UAT sign-off, immutability enforced
10	Scale Validated	1,500 readings/sec sustained, graceful degradation verified
11	Production Ready	All UAT sign-offs, zero P1/P2 bugs, deployment rehearsed
12	Go-Live	Zero downtime deployment, first 1M readings processed

How We Ensure Delivery

Continuous Stakeholder Alignment

- Weekly sprint reviews with Product Owner and business stakeholders
- Bi-weekly demos to supervisors and field techs (Weeks 5-11)
- Daily stand-ups (15 min) with engineering team

Managing Dependencies Early

- Infrastructure pre-provisioned in Week 0 (before sprint 1)
- SSO integration tracked weekly with Identity Team
- Billing team sync every Monday (Weeks 8-11)

Iterative Delivery (Agile Mindset)

- 2-week sprints with deployable increments
- UAT feedback incorporated within sprint (not deferred)
- Retrospectives every 2 weeks to adjust course

Tracking Success via Metrics

- Daily monitoring of throughput, latency, error rates

- Weekly review of UAT feedback scores
- Sprint velocity tracked (planned vs delivered story points)

Budget Summary

Category	Amount	% of Total
Team costs (7.5 FTE × 3 months)	R 2,310,000	85%
Infrastructure (database, compute, network)	R 159,000	6%
Tools & training	R 15,000	1%
Contingency (10% buffer)	R 248,400	9%
Total	**R 2,732,400**	**100%**

Designs:

Roadmap timeline

Reading Validation Component - Q2 2026 Roadmap												
Timeline	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Sprint	Sprint 1	Sprint 1	Sprint 2	Sprint 2	Sprint 3	Sprint 3	Sprint 4	Sprint 4	Sprint 5	Sprint 5	Sprint 6	Launch
Workstream / Activity												
DATABASE												
Schema Design	BUILD	BUILD										
Query Optimization			BUILD	BUILD								
Performance Tuning					BUILD	BUILD						
Load Testing									BUILD	BUILD		
BACKEND												
API Framework	SETUP	SETUP										
Auto Validation			BUILD	BUILD								
Manual Approval APIs					BUILD	BUILD						
Correction Flow APIs							BUILD	BUILD				
Integration Testing											UAT	
FRONTEND												
Component Library			SETUP	SETUP								
Supervisor Dashboard					BUILD	BUILD						
Field Tech UI							BUILD	BUILD				
UI Polish									BUILD	BUILD		
DEVOPS												
Infrastructure Setup	PROVISION	PROVISION										
Monitoring Setup			CONFIG	CONFIG								
CI/CD Pipeline			DEPLOY	DEPLOY								
Performance Testing									TEST	TEST		
Production Deployment												DEPLOY
MILESTONES												
Schema Complete		✓										
Auto Validation Works				✓								
Manual Approval Live						✓						
Correction Complete								✓				
Scale Validated										✓		
Production Ready											✓	
Go-Live												✓

Roadmap

Component	Owner	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Database Schema	DBA												
API Framework	Backend												
Auto Validation	Backend												
Supervisor UI	Frontend												
Field Tech UI	Frontend												
Correction Flow	Backend												
Performance Testing	DevOps												
Integration	All												
Production Deploy	DevOps												
MILESTONES													
Schema Complete			✓										
Auto Validation					✓								
Manual Approval							✓						
Correction Complete								✓					
Scale Validated										✓			
Production Ready											✓		
Go-Live													✓

Budget

BUDGET BREAKDOWN - Q2 2026 (12 Weeks)				
Category	**Line Item**	**Monthly Cost**	**Q2 Total (3 mor**	**% of Total**
TEAM COSTS				
Lead Systems Analyst	Design & coordination	R 120,000	R 360,000	13%
Product Owner (0.5 FTE)	Backlog management	R 60,000	R 180,000	7%
Tech Lead	Architecture	R 110,000	R 330,000	12%
Backend Engineers (2)	Development	R 160,000	R 480,000	18%
Frontend Engineer	UI development	R 80,000	R 240,000	9%
DBA	Schema & performance	R 90,000	R 270,000	10%
QA Engineer	Testing	R 70,000	R 210,000	8%
DevOps Engineer	Infrastructure	R 80,000	R 240,000	9%
TEAM SUBTOTAL		R 770,000	R 2,310,000	85%
INFRASTRUCTURE COSTS				
PostgreSQL + TimescaleDB	Database cluster	R 30,000	R 90,000	3%
Application servers	Kubernetes (3 AZs)	R 15,000	R 45,000	2%
Network	Load balancers, VPC	R 5,000	R 15,000	1%
Monitoring	Grafana, Prometheus	R 3,000	R 9,000	0%
INFRASTRUCTURE SUBTOTAL		R 53,000	R 159,000	6%
TOOLS & TRAINING				
Load testing tools	Locust, k6 licenses	-	R 10,000	0%
Training materials	Documentation, videos	-	R 5,000	0%
TOOLS SUBTOTAL		-	R 15,000	1%
CONTINGENCY				
10% buffer	Risk mitigation	-	R 248,400	9%
CONTINGENCY SUBTOTAL		-	R 248,400	9%
TOTAL BUDGET		R 823,000	R 2,732,400	100%

Summary

This plan delivers a production-ready reading validation component in 12 weeks by prioritizing revenue protection (48M readings/day with zero data loss), automating 95% of volume (reducing manual work), and validating scale before go-live (1,500 readings/sec tested).

The sequence is deliberate: foundation first (correct schema prevents expensive rework), automated before manual (95% volume is higher priority), and performance testing before deployment (validate capacity before risking production). Success is measured by transaction accuracy (≥99.9%), throughput (1,500 readings/sec), and operational adoption (90% trained users).